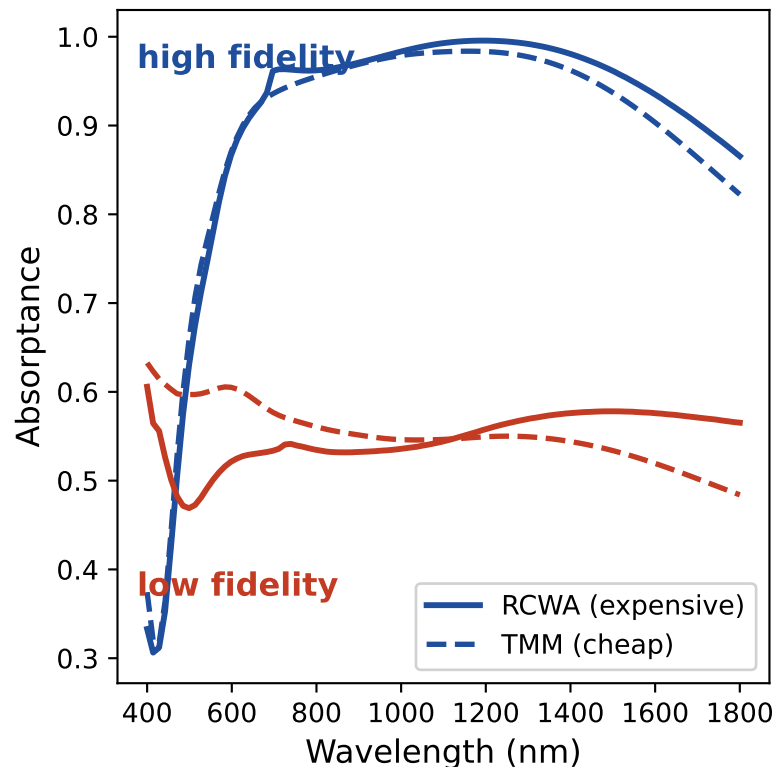


Will cheap-physics pre-training help this structure? A 50-sample pilot answers before the 4-5 h pipeline.

1. Run a 50-sample RCWA pilot (~ 15-30 min)



2. Compute the joint r -MAE diagnostic

shape: median per-sample
Pearson r (TMM, RCWA)

amplitude: median
operating-band MAE

*no PCA · no pre-trained model
raw simulator outputs only*

3. Decide before training (benefit tracks MAE, $|\text{Pearson}| = 0.98$)

